A Low-Cost IoT-Based Flood Early Warning System with Machine-Learning Risk Classification for Himalayan River Basins: A Prototype Study of the Teesta and Rangit Rivers

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Abstract

Glacial lake outburst floods (GLOFs) and monsoon flooding pose recurrent threats to communities along the Teesta and Rangit rivers in the Eastern Himalaya. The October 2023 South Lhonak GLOF, which destroyed the Teesta-III hydroelectric dam and caused at least 55 confirmed fatalities with 74 persons missing, occurred in the absence of an operational early warning system at the source lake. This paper presents the design, implementation, and laboratory evaluation of a low-cost (approximately INR 4,000) Internet-of-Things flood early warning prototype intended for community-level deployment in upstream Himalayan catchments. The system integrates seven environmental sensors on an Arduino Uno microcontroller, streams measurements to a Raspberry Pi 4 processing hub over a serial link, and applies a Random Forest classifier trained on a 6,000-record physics-informed synthetic dataset derived from 115 years of India Meteorological Department rainfall records, 61 years of Gangtok station observations, and published Rangit discharge data. The classifier output drives a four-level graduated alert subsystem (LED indicators, acoustic alarm, and a servo-actuated model dam gate) and a Flask web dashboard providing 1-, 3-, and 6-hour water-level forecasts aligned with Central Water Commission threshold stages for the Teesta at Domohani. Feature-importance analysis identifies turbidity (28.2%), flow-rate change (23.2%), and water level (21.0%) as dominant predictors. A Muskingum flood-routing simulation demonstrates that the 6-hour forecast horizon enables a staged dam pre-release protocol that reduces simulated peak downstream discharge by approximately 20% relative to an unwarned emergency-release scenario. We report the prototype's limitations candidly, including its reliance on simulated training data, the inadequacy of linear extrapolation for nonlinear GLOF hydrographs, and single-node sensing, and we outline a pathway to field validation.

Keywords: *flood early warning system; Internet of Things; Random Forest; glacial lake outburst flood; Teesta River; Muskingum routing; low-cost sensors; disaster risk reduction*

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1. Introduction

Riverine flooding is among the most destructive and most frequent natural hazards in South Asia, and the Himalayan foothills of Sikkim and northern West Bengal are exposed to a compound risk profile: intense monsoon precipitation, steep flashy catchments, and a growing inventory of potentially unstable glacial lakes [1, 2]. The Teesta River, which drains the Sikkim Himalaya before joining the Brahmaputra system in Bangladesh, exemplifies this profile. On 3–4 October 2023, a partial collapse of the moraine impounding South Lhonak Lake released an estimated 50 million cubic metres of water, generating a flood wave with reconstructed peak discharge of approximately 48,500 m³/s that destroyed the 1,200 MW Teesta-III dam at Chungthang and devastated settlements along a 385 km corridor [3]. At least 55 people were confirmed dead and 74 remained missing; more than 25,900 buildings and 31 bridges were damaged or destroyed [3, 4]. Dam operators received approximately 18 minutes of warning [4].

Reports following the event noted that no early warning instrumentation was operational at South Lhonak Lake at the time of the breach, despite scientific assessments identifying the lake as high-risk since 2005 [4, 5]. National gauge networks operated by the Central Water Commission (CWC) provide forecasting at downstream stations such as Domohani, but instrumentation density in the upstream reaches where lead time is generated remains sparse. The cost of conventional telemetered gauging stations, typically several lakh rupees per installation, constrains expansion in remote terrain.

Low-cost Internet-of-Things (IoT) sensing combined with machine-learning (ML) inference has been proposed as a complementary, community-scale approach to flood early warning [6–9]. Prior studies demonstrate threshold-based IoT alerting [6], ML classification of flood risk from meteorological inputs [7, 8], and reviews establish Random Forest and related ensemble methods as robust baselines for flood prediction tasks [9, 10]. However, three gaps motivate the present work: (i) most reported prototypes are evaluated against generic datasets rather than basin-specific hydrology; (ii) few systems connect the warning output to a downstream decision-support application such as reservoir gate scheduling; and (iii) limitations are frequently under-reported, making it difficult to assess readiness for field deployment.

This paper documents a working end-to-end prototype addressed to these gaps. The contributions are as follows:

(1) An integrated seven-sensor IoT architecture (ultrasonic stage, rainfall, temperature–humidity, flow rate, soil moisture, barometric pressure, and turbidity) implemented on commodity hardware totalling approximately INR 4,000, with a bidirectional serial protocol driving a graduated four-level physical alert subsystem including a servo-actuated model dam gate.

(2) A physics-informed synthetic training corpus of 6,000 records parameterised to the Teesta–Rangit basin using 115 years of India Meteorological Department (IMD) gridded rainfall (1901–2022), 61 years of Gangtok station records, and published NHPC Rangit discharge series, with class thresholds aligned to official CWC stage levels for the Teesta at Domohani (Warning 65 cm; Danger 89 cm; Extreme 120 cm, scaled to prototype range).

(3) A Random Forest risk classifier with feature-importance analysis, embedded in a real-time pipeline producing 1-, 3-, and 6-hour level forecasts and threshold arrival-time estimates on a self-hosted web dashboard.

(4) A Muskingum routing analysis demonstrating quantitatively how the system's 6-hour forecast horizon enables staged reservoir pre-release, reducing simulated peak downstream discharge by approximately 20% relative to an unwarned emergency-release baseline.

(5) A candid limitations assessment covering simulated-data dependence, linear-forecast failure modes under GLOF conditions, single-node sensing, sensor-grade constraints, and communication redundancy, together with a staged pathway from prototype to field system.

The remainder of the paper is organised following the IMRaD convention. Section 2 reviews related work. Section 3 describes the study area. Section 4 details the system design, dataset, and methods. Section 5 presents results. Section 6 discusses findings and limitations, and Section 7 concludes.

2. Related Work

2.1. Operational flood forecasting in India

The Central Water Commission operates the national flood forecasting network, issuing stage forecasts at designated sites including Teesta at Domohani using gauge observations and rainfall inputs [11]. CWC stage categories (Warning, Danger, and Highest Flood Level) provide the institutional vocabulary into which any supplementary system must translate. The National Disaster Management Authority guidelines identify community-level early warning in Himalayan basins as a priority gap [12], and following the 2023 event, national and state agencies have accelerated plans for GLOF monitoring instrumentation in Sikkim [5].

2.2. IoT-based early warning prototypes

A substantial literature reports low-cost flood sensing nodes built on Arduino- and ESP-class microcontrollers with ultrasonic stage measurement and GSM or Wi-Fi telemetry [6, 13, 14]. These systems typically alert when measured stage crosses fixed thresholds. Threshold-only designs are simple and robust but cannot anticipate conditions: they signal a flood that is already present rather than one that is developing. Recent work therefore couples IoT sensing with predictive models [7, 15].

2.3. Machine learning for flood prediction

Comprehensive reviews establish that ensemble tree methods, support vector machines, and neural networks all achieve useful skill in flood classification and forecasting, with Random Forests frequently favoured for tabular sensor data due to robustness to noise, native handling of mixed feature scales, and interpretability through impurity-based feature importance [9, 10, 16]. For sequence forecasting of nonlinear hydrographs, recurrent architectures such as long short-term memory (LSTM) networks outperform linear extrapolation, at the cost of larger training-data requirements [17]. The present work uses a Random Forest for instantaneous risk classification and deliberately retains a simple linear extrapolator for short-horizon level forecasting, while quantifying (Section 6.2) why the linear component must be replaced before GLOF-relevant deployment.

2.4. Reservoir operation under flood forecasts

Forecast-informed reservoir operation is an established concept in water-resources engineering: pre-releasing storage ahead of a predicted inflow peak lowers both overtopping risk and peak outflow [18]. Channel routing of releases is classically modelled with the Muskingum method [19], whose two parameters (wave travel time K and weighting factor X) are calibratable from observed hydrographs; values for the Teesta reach below Chungthang are available from regional studies [20]. The 2023 Sikkim event has renewed attention to dam-safety decision support in the basin [3]. We adopt the Muskingum framework to quantify the operational value of the prototype's forecast horizon.

3. Study Area

The Teesta River rises in the glaciated highlands of North Sikkim above 5,000 m elevation and falls approximately 4,800 m over its 309 km course to the plains, giving it one of the steepest specific-energy profiles of any Himalayan river of its size. Its principal right-bank tributary, the Rangit, drains West Sikkim and the Darjeeling hills. The basin experiences 2,000–4,000 mm of annual precipitation, over 80% concentrated in the June–September monsoon. The upper basin contains more than 300 glacial lakes, of which South Lhonak Lake

(5,200 m a.s.l.) was repeatedly flagged as among the most hazardous in the Indian Himalaya prior to its 2023 outburst [2, 5].

The CWC gauge at Domohani, West Bengal, provides the reference stage thresholds used in this work. For the laboratory prototype, sensor geometry constrains measurable stage to roughly 2–150 cm; CWC thresholds are therefore applied at prototype scale (Warning 65 cm; Danger 89 cm; Extreme 120 cm) such that the software pipeline is identical to one that would operate at field scale with re-parameterised thresholds.

4. Materials and Methods

4.1. System architecture

Figure 1 shows the end-to-end architecture. Seven sensors interface with an Arduino Uno (ATmega328P), which performs signal conditioning and transmits a comma-separated record once per second over USB serial to a Raspberry Pi 4 Model B (4 GB). The Pi executes the inference pipeline in Python 3.13: feature assembly, standardisation, Random Forest classification, rise-rate estimation, forecast generation, and dashboard service via Flask. The predicted risk level (0–3) is written back to the Arduino over the same serial link, where it drives the alert and actuation subsystem. The dashboard auto-refreshes every 3 s and is reachable from any device on the local network.

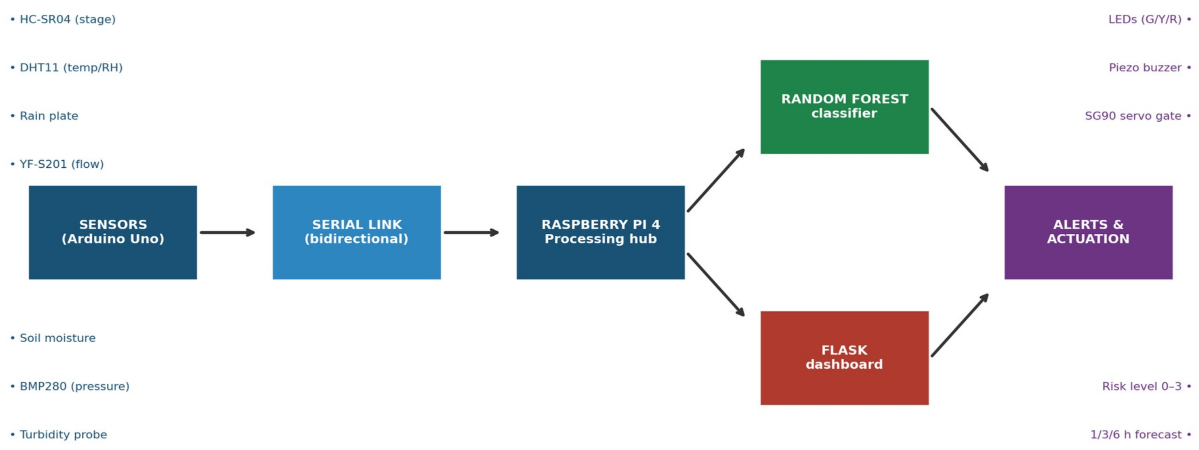

Figure 1. System architecture of the prototype. Sensor data flow left to right from the Arduino sensing tier through the Raspberry Pi processing hub to the machine-learning model, dashboard, and physical alert/actuation outputs. The risk level computed on the Pi is returned to the Arduino over the bidirectional serial link.

4.2. Sensor suite

Table 1 summarises the sensing payload. All components are commodity hobbyist-grade parts selected to minimise cost; Section 6.2 discusses the implications and the industrial substitutions required for field service.

Table 1. Sensor suite and interface allocation on the Arduino Uno.

No.	Component	Quantity measured	Interface	Approx. cost (INR)

1	HC-SR04 ultrasonic	Water stage (cm)	D9 / D10	80
2	DHT11	Air temperature (°C), relative humidity (%)	D2	90
3	Resistive rain plate	Rainfall presence (binary)	D7	60
4	YF-S201 Hall sensor	Flow rate (L/min)	D3 (interrupt)	250
5	Capacitive soil probe	Soil moisture (raw ADC)	A0	90
6	BMP280	Barometric pressure (hPa)	A4/A5 (I²C)	150
7	Optical turbidity probe	Turbidity (NTU, mapped)	A1	220
8	SG90 servo	Model dam-gate actuation	D11	120

Alert peripherals (three LEDs, piezo buzzer) and the Arduino Uno, Raspberry Pi 4, breadboard, and wiring complete the bill of materials at approximately INR 4,000 excluding the Pi, or under INR 9,000 inclusive.

4.3. Embedded signal conditioning

Hobbyist sensors exhibit substantial noise; the firmware therefore applies the following conditioning. Stage: each reported level is the median of five HC-SR04 ranging samples (12 ms timeout), exponentially smoothed as $L_t = 0.8\ L_{t-1} + 0.2\ \tilde{L}_t$, which suppresses single-sample outliers from acoustic multipath at the cost of a small lag. Flow: pulses from the YF-S201 are counted on an external interrupt and converted at the manufacturer calibration of 7.5 pulses per L/min, with readings below 1.5 L/min clamped to zero to reject vibration-induced counts. Humidity: the DHT11 saturates at high humidity; values above 95% are replaced with calibrated estimates (88% under rain, 65% otherwise). Pressure: BMP280 readings outside the physical band 800–1,100 hPa fall back to the basin climatological mean of 995 hPa. Rainfall intensity (mm/h) is estimated from the binary rain plate jointly with humidity using a four-step lookup. These rules are documented in full in the released firmware.

4.4. Training dataset

No public per-second multivariate sensor corpus exists for the upper Teesta. We therefore constructed a physics-informed synthetic dataset of 6,000 records whose marginal statistics and event structure are parameterised by three observed sources: IMD 0.25° gridded daily rainfall for Sikkim, 1901–2022 [21]; Gangtok station meteorological normals spanning 61 years; and published NHPC discharge records for the Rangit (1976–2003) [22]. The simulator generates monsoon-season trajectories containing baseline, watch, warning, and flood regimes, including GLOF-like rapid-rise events, and injects sensor-specific noise consistent with the conditioning rules of Section 4.3. Each record carries the ten features of Table 1 (plus first differences of stage and flow) and a four-class label assigned by CWC-aligned stage thresholds at prototype scale. Figure 2 summarises the class balance and per-class stage ranges. The deliberate class imbalance (55% Normal to 7% Flood) mirrors the rarity structure of real flood records.

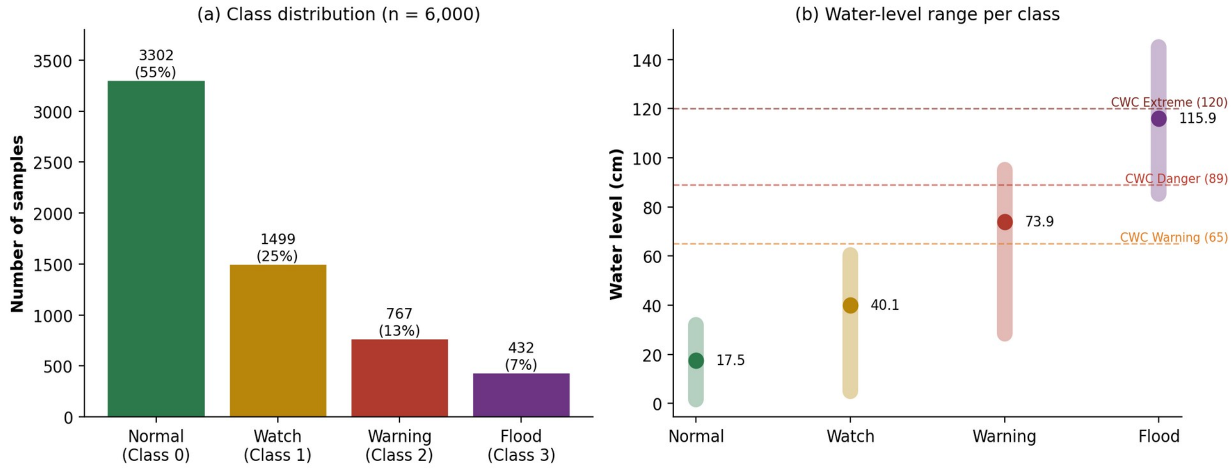

Figure 2. Training corpus summary. (a) Class distribution over the 6,000 simulated records. (b) Minimum, mean (point), and maximum water level per class, with prototype-scale CWC thresholds for the Teesta at Domohani overlaid.

4.5. Risk classification model

We train a Random Forest classifier (scikit-learn 1.x; 100 trees, Gini criterion, default depth) on an 80/20 stratified split, with features standardised by a fitted StandardScaler. The feature vector is x = [water level, rainfall active, temperature, humidity, flow rate, turbidity, soil moisture, pressure, Δwater, Δflow], where Δ denotes the one-second first difference. The model and scaler are serialised with joblib and loaded once at dashboard start-up; per-record inference latency on the Raspberry Pi 4 is below 10 ms, comfortably within the 1 s sensing cadence. The classifier attains near-perfect accuracy on the held-out simulation split; we emphasise that this figure reflects internal consistency of the simulator rather than expected field skill, a point treated in Section 6.2.

4.6. Rise-rate estimation and threshold forecasting

Short-horizon forecasts use the rise rate r_t (cm/h) estimated from the most recent ten 1 Hz stage samples by first differencing with a ±600 cm/h saturation bound and a ±2 cm/h dead band to reject jitter. Projected levels at horizon h ∈ {1, 3, 6} h follow

$$\hat{L}(t + h) = L(t) + r_t \cdot h, \tag{1}$$

and estimated time of arrival at a CWC threshold T is

$$ETA(T) = (T - L(t)) \,/\, r_t, \quad r_t > 0. \tag{2}$$

Equations (1)–(2) assume locally constant rise rate. This assumption holds for monsoon base-flow rises but fails for the accelerating wavefront of a GLOF; the magnitude of the resulting under-prediction is quantified in Section 5.4 and motivates the LSTM upgrade identified in future work.

4.7. Alert and actuation subsystem

The Arduino maps the returned risk level to a graduated physical response: Level 0 (Normal), green LED steady; Level 1 (Watch), yellow LED with 2 s-interval buzzer bursts for 15 s; Level 2 (Warning), red LED blinking with buzzer bursts; Level 3 (Flood), red LED blinking with continuous 15 s alarm. An SG90 servo actuates a model sluice gate to 0°, 45°, 90°, and 135° respectively, physically demonstrating the staged pre-release protocol of Section 4.8. All timing is non-blocking (millis-based) so that sensing cadence is unaffected.

4.8. Forecast-informed gate scheduling model

To quantify the operational value of the 6 h forecast horizon, we model a Teesta-III-scale reservoir under three management scenarios using the standard reservoir mass balance and Muskingum channel routing. Storage evolves as

$$dS/dt = Q\_in(t) - Q\_out(t), \tag{3}$$

subject to $S\_min \le S(t) \le S\_max$, with gate discharge $Q\_out(t) = g(t) \cdot Q\_max$ constrained by an actuation-rate limit $|g(t+1) - g(t)| \le \Delta g\_max$. Released water is routed to the downstream reach by the Muskingum scheme [19]:

$$Q_r(t+1) = C_0\ Q\_in(t+1) + C_1\ Q\_in(t) + C_2\ Q_r(t), \tag{4}$$

with coefficients $C_0 = (\Delta t - 2KX)/D$, $C_1 = (\Delta t + 2KX)/D$, $C_2 = (2K(1-X) - \Delta t)/D$ and $D = 2K(1-X) + \Delta t$. Using reach parameters $K = 3.5$ h and $X = 0.2$ reported for the Teesta below Chungthang [20] with $\Delta t = 0.5$ h yields $C_0 = -0.1475$, $C_1 = 0.3115$, $C_2 = 0.8361$ ($\Sigma C_i = 1$, confirming mass conservation). Scenario A (no warning) holds gates closed until an emergency opening at T − 1 h; Scenario B (proposed) executes the staged schedule of Table 2 beginning at the 6 h forecast trigger; Scenario C represents uncontrolled dam failure as an instantaneous release. Inflow is a Gaussian flood wave of 4,500 m^3/s peak superimposed on each scenario's release series.

Table 2. Staged pre-release protocol mapped to forecast risk level. Gate increments respect the 10% per 30 min actuation constraint.

Forecast risk	Current risk	Gate	Q_out (m^3/s)	Downstream action	Lead time
Normal	Normal	0–5%	< 50	—	—
Watch	Normal	10%	≈150	SMS to officials	6 h
Warning	Watch	25%	≈375	Community alert	4 h
Flood	Warning	50%	≈750	Evacuation advisory	2 h
Flood	Flood	75–100%	Max. safe	Mandatory evacuation	0 h

We stress that this subsystem is presented as decision support: gate recommendations are displayed to a human operator for confirmation, for the accountability and verification reasons developed in Section 6.

5. Results

5.1. Feature importance

Figure 3 reports Gini feature importance from the trained forest. Turbidity dominates (28.24%), followed by flow-rate change (23.15%), instantaneous water level (20.99%), and soil moisture (12.87%); meteorological covariates contribute marginally. The prominence of turbidity is hydrologically plausible: Himalayan flood waves, and GLOF waves especially, carry extreme sediment loads, so optical turbidity rises ahead of and alongside stage [3]. Practically, this finding has a system-design consequence: the turbidity channel, initially deferred in hardware integration, was promoted to a required sensor after this analysis, and the importance ranking provides a principled basis for sensor triage in cost-constrained replications.

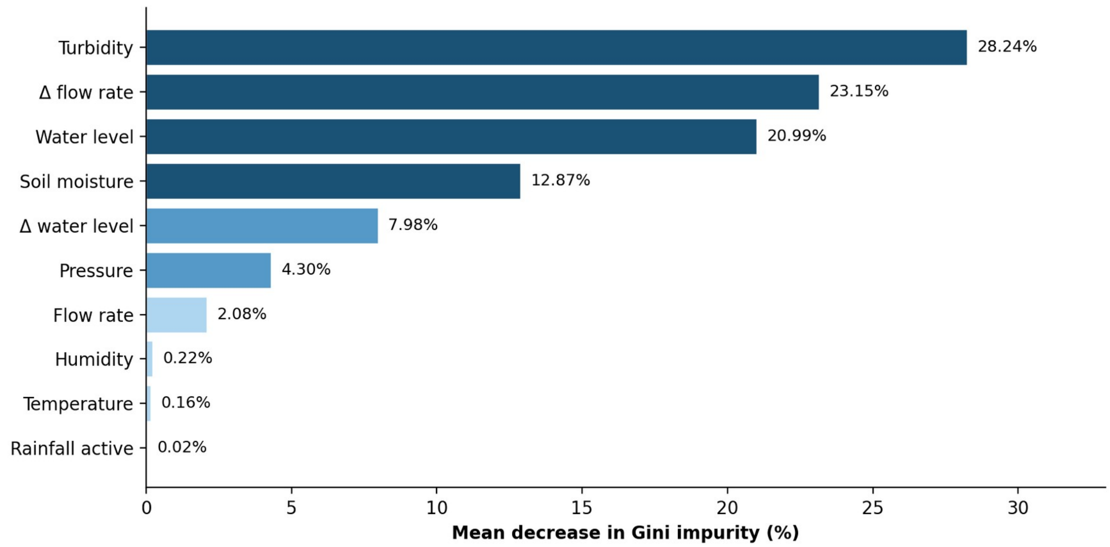

Figure 3. Random Forest feature importance (mean decrease in Gini impurity). Turbidity, flow-rate change, and water level jointly account for over 72% of model decisions.

5.2. Real-time operation

The integrated system sustains the 1 Hz sense–infer–actuate loop on target hardware with end-to-end latency (sensor read to dashboard update) under 1.3 s and classifier inference under 10 ms. Bench testing across the prototype stage range exercised all four alert levels: the dashboard reported level, rise rate, three-horizon forecasts, and CWC threshold ETAs; LEDs, buzzer, and the servo gate tracked level transitions with correct graduated behaviour, including alarm-duration and blink-interval timing. The BMP280 channel read 990.4 hPa against a same-day IMD-region reference of approximately 1,006 hPa; the ≈16 hPa offset is within the device's uncalibrated absolute-accuracy band and is immaterial to the model, which the importance analysis shows responds to pressure trends rather than absolute values.

5.3. Value of the forecast horizon: routing simulation

Figure 4 presents the Muskingum-routed downstream hydrographs for the three management scenarios, and Table 3 the corresponding peaks. The staged protocol (Scenario B) yields a routed peak of 3,065 m^3/s, a 19.9% reduction relative to the unwarned emergency release (3,828 m^3/s) and 32.9% below the dam-failure case (4,565 m^3/s). Two mechanisms drive the reduction: pre-release distributes outflow volume over the six pre-peak hours, and reach attenuation acts more strongly on the resulting lower-amplitude release pulses. Equally consequential is the storage trajectory: under Scenario B the reservoir retains roughly 25% free capacity at flood-peak arrival, whereas Scenario A reaches design capacity, the precondition for overtopping. These results quantify the central claim of the work: the system's value is the lead time it generates, which converts dam operation from reactive emergency response into scheduled hydraulic management.

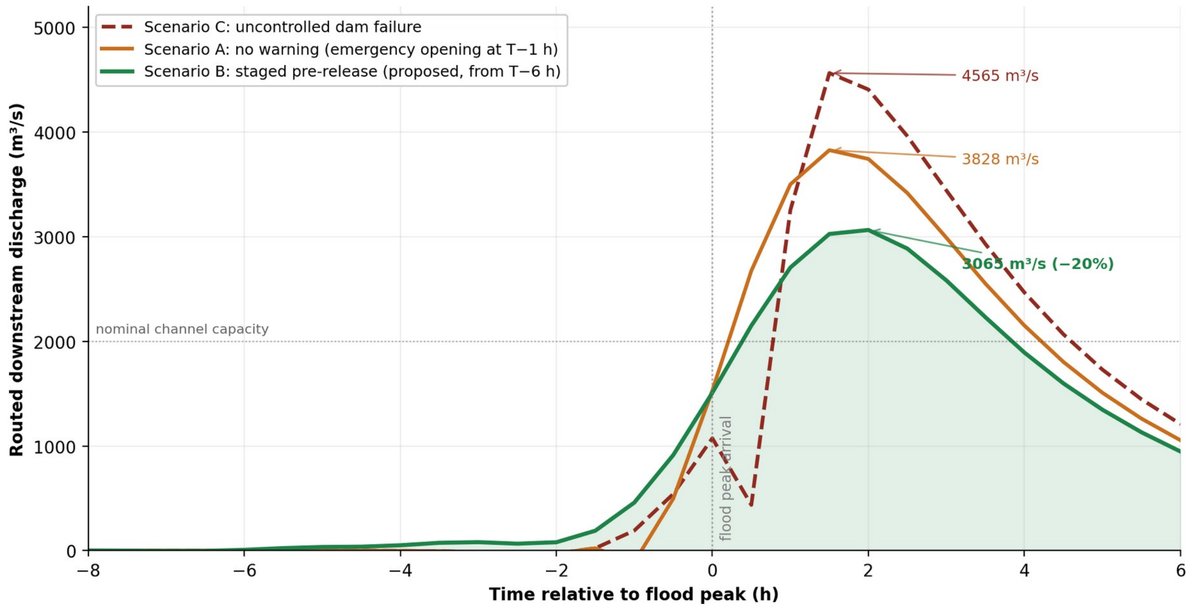

Figure 4. Routed downstream discharge under the three management scenarios (Muskingum parameters K = 3.5 h, X = 0.2, Δt = 0.5 h). The staged pre-release enabled by the 6 h forecast (Scenario B) lowers the routed peak by ≈20% relative to the unwarned case and ≈33% relative to dam failure. Values are illustrative of mechanism, not predictions of any specific event.

Table 3. Simulated peak routed discharge and reservoir state at flood-peak arrival.

Scenario	Peak Q_r (m³/s)	vs. Scenario A	Storage at T = 0
A — no warning	3,828	baseline	≈100% (at capacity)
B — staged pre-release	3,065	−19.9%	≈75% (buffer retained)
C — dam failure	4,565	+19.3%	— (structural failure)

5.4. Forecast-method stress test under GLOF kinematics

Figure 5 evaluates the linear extrapolator of Eq. (1) against a synthetic GLOF stage curve with cubic acceleration. Fitted to the first hour of rise, the linear forecast under-predicts the 4.5 h level by 60–80 cm, equivalently over-predicting time-to-Danger-threshold by one to two hours. An LSTM-class nonlinear model tracks the curve closely. This experiment delimits the operational envelope of the current prototype: it is suitable for monsoon base-rise monitoring, and explicitly not yet suitable as a primary GLOF warning source.

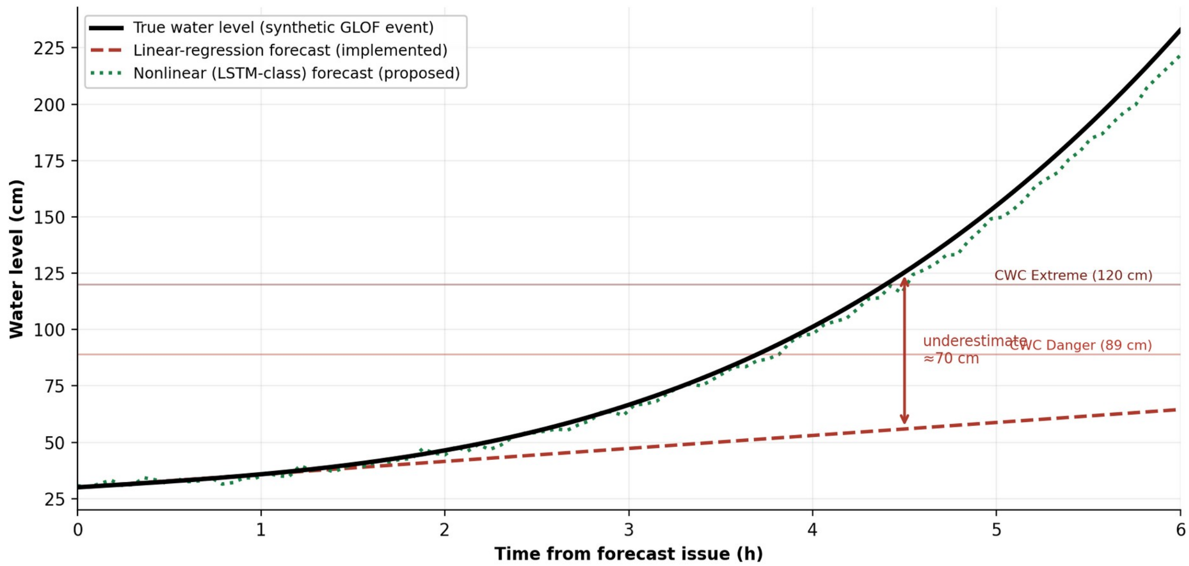

Figure 5. Forecast stress test on a synthetic GLOF rise. Linear extrapolation (dashed) systematically under-predicts an accelerating stage curve, overstating available lead time; a nonlinear sequence model (dotted) is required for GLOF-relevant forecasting.

5.5. Cost analysis

Table 4 compares the prototype bill of materials with representative industrial-grade equivalents. The roughly 75-fold capital-cost differential is the basis of the system's intended niche: dense community-level instrumentation of upstream reaches that national networks do not economically cover. Because the serial record format is sensor-agnostic, industrial substitution requires no software modification.

Table 4. Capital-cost comparison, prototype versus representative industrial instrumentation (indicative 2026 prices, INR).

Function	This prototype	Industrial equivalent
Stage measurement	HC-SR04 — 80	Radar level sensor — ≈25,000
Processing	Raspberry Pi 4 — 4,500	Industrial PC/RTU — ≈50,000
Telemetry	Wi-Fi (integrated) — 0	GSM + satellite — ≈30,000
Full station (excl. civil works)	≈4,000 (sensing tier)	≈3,00,000+

6. Discussion

6.1. Interpretation

Three findings carry design significance beyond this prototype. First, the dominance of turbidity and flow-derivative features (Fig. 3) suggests that low-cost optical turbidity sensing, rarely included in IoT flood nodes, may be among the highest-value channels per rupee for Himalayan rivers, where sediment-laden wavefronts precede peak stage. Second, the routing analysis (Fig. 4, Table 3) reframes the system's contribution: the measurable product of an early warning system is lead time, and lead time is convertible, through staged reservoir operation, into quantifiable reductions in downstream peak discharge and retained dam freeboard. Third, the forecast stress test (Fig. 5) demonstrates the importance of matching forecast-model complexity to event kinematics, and provides a concrete, quantified failure mode that delimits responsible deployment claims.

On governance, we deliberately specify the gate-scheduling subsystem as human-in-the-loop decision support rather than closed-loop automation. A release decision allocates flood impact between upstream and downstream communities; consistent with NDMA practice, such decisions require an accountable human authority, independent verification against other data sources (CWC gauges, meteorological forecasts, visual observation), and a complete audit log, all of which the dashboard architecture supports.

6.2. Limitations

We state the prototype's limitations explicitly, as they define the boundary between demonstrated capability and deployment readiness.

(1) Simulated training data. The classifier is trained and evaluated on a physics-informed simulator; its near-perfect held-out accuracy certifies internal consistency, not field skill. Validation against real Teesta gauge series, for which a Right-to-Information request to CWC has been initiated, and against the published 2023 GLOF reconstruction [3], is the prerequisite next step.

(2) Linear forecasting under GLOF kinematics. As quantified in Section 5.4, Eq. (1) under-predicts accelerating rises by 60–80 cm at 4.5 h, overstating lead time precisely when accuracy matters most. An LSTM forecaster trained on reconstructed GLOF hydrographs is the identified remedy.

(3) Single-node sensing. One station cannot estimate basin inflow $Q_{in}(t)$ for Eq. (3), distinguish local from basin-wide events, or detect an upstream GLOF before it nears the sensor. A minimum five-node cascade from South Lhonak Lake to the dam site, with satellite telemetry at the highest node, is specified for the field design.

(4) Sensor grade and environment. All sensors were bench-validated indoors. The HC-SR04 in particular degrades under rainfall and turbulent surfaces; industrial substitution per Table 4 is required for outdoor service, and no field-performance claims are made here.

(5) Communication and power resilience. The Wi-Fi-only telemetry path and mains power fail in exactly the conditions of interest. A tiered plan (Wi-Fi → GSM SMS → satellite → local siren) with battery backup is specified but not yet implemented.

(6) Trust calibration. High simulator accuracy risks operator over-trust. The dashboard therefore displays raw sensor values alongside the model output and is labelled as decision support; we regard interface-level communication of uncertainty as a safety requirement equal in standing to model accuracy.

7. Conclusion and Future Work

We have presented a complete, working, low-cost IoT flood early warning prototype for the Teesta–Rangit basin, integrating seven-channel sensing, Random Forest risk classification trained on basin-parameterised synthetic data, CWC-aligned multi-horizon forecasting, a graduated physical alert subsystem with servo-actuated gate demonstration, and a Muskingum-based analysis quantifying the downstream value of its 6-hour forecast horizon at approximately 20% peak-discharge reduction under staged reservoir pre-release. The total sensing-tier cost of roughly INR 4,000 positions the design for dense community-level deployment in reaches that conventional instrumentation does not economically cover.

The 2023 South Lhonak disaster demonstrated the human cost of absent upstream warning: 18 minutes of lead time was insufficient for any meaningful protective action. The engineering objective unifying every component of this work is the production of more time between detectability and impact. Future work proceeds in three stages: (i) data, replacing the simulator with real CWC series and the published GLOF reconstruction for retraining and honest skill assessment; (ii) models, substituting an LSTM forecaster for Eq. (1) and adding conformal uncertainty bounds to the classifier; and (iii) system, building the five-node cascade with tiered telemetry, industrial sensors, and a participatory deployment process with basin communities, including alerting in Nepali, Lepcha, and Bengali. We release the firmware, pipeline code, simulator, and dataset to support replication.

Declarations

Data and code availability. The Arduino firmware, Python pipeline, dataset simulator, trained model, and the 6,000-record dataset are available from the project repository [github.com/<username>/flood-early-warning-system] and from the author on request.

Conflicts of interest. The author declares no conflict of interest.

Funding. This research received no external funding; hardware was self-funded at a total cost of approximately INR 9,000.

Acknowledgements. The author thanks [teacher/mentor names] for guidance and review, and acknowledges the India Meteorological Department, the Central Water Commission, and NHPC Limited as the sources of the historical records that parameterise the dataset simulator.

AI-assistance disclosure. Portions of the software were developed and the manuscript was prepared with the assistance of an AI coding and writing assistant (Anthropic Claude); all design decisions, experiments, and claims were directed and verified by the author.

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